Audio Emotion Classifier

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Intro

Why Emotion in Speech Matters

- Humans communicate emotion through voice
- Current AI understands what we say, not how we feel
- The ability to automatically recognize emotions has become increasingly important:
    - human-computer interaction (AI companionship)
    - virtual assistants
    - mental health assessment

Challenge of Emotion Recognition

- Speech varies by:
  - Speaker
  - Language
  - Recording conditions
- Models trained on one dataset may not perform well in real-world data

👉 **Generalization is the main problem.**

Research Objective

- ★ The **goal**: to test whether a single model can learn general emotional patterns across diverse data
- ★ Approach:
    - Combine multiple datasets
    - Employ a hybrid deep learning model

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Dataset

Multi-Dataset

- 6 emotions:
  - Angry, Disgust, Fear, Happy, Neutral, Sad

- 10,000+ audio samples from 9 datasets
  - Public datasets: RAVDESS, CREMA-D, EmoDB, ShEMO…
  - Self-collected: Friends, iPartment(Chinese TV series)

★ By combining datasets, I am trying to simulate more realistic conditions and test how well the model generalizes

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Preprocessing

Preprocessing Pipeline

- From Audio → Model Input
- Steps:
  - Resample to 22,050 Hz
  - Trim silence
  - Fixed length (8 sec)

⇨ Convert to Mel-spectrogram

- Data Augmentation
  - Time masking + Frequency masking (randomly hide parts of the spectrogram)

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Model

Model Architecture

Mel-Spectrogram

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ConvNeXt CNN Backbone

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Transformer Encoder

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Attention Pooling

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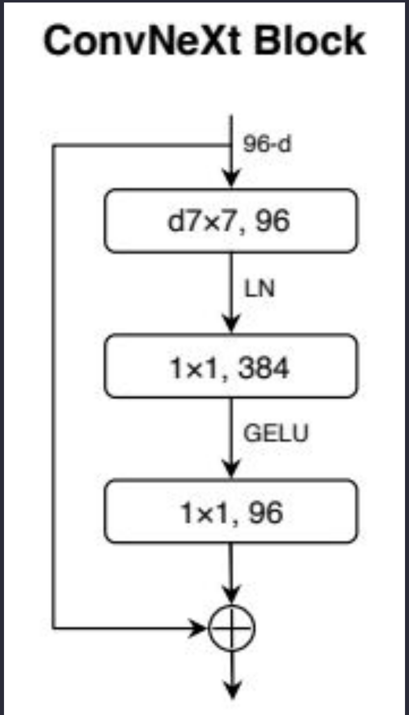
Emotion Prediction

- Hybrid CNN–Transformer Model
- Pipeline:
  - CNN → extracts local features
  - Transformer → capture time relationships
  - Attention → focus on important parts

For ConvNeXt…

- Outside ConvNeXt blocks:
    - A stem convolution layer that increases the channel dimension to 32
- Inside:
    - A depthwise convolution (kernel 7×7) to process channels independently
    - A normalization layer to stabilize feature distributions
    - A pointwise convolution-based MLP that expands and then compresses channel dimensions
    - A residual connection to facilitate gradient flow



For Transformer + Attention…

- Transformer:
  - Transformer encoder composed of two layers with multi-head self-attention
  - Capture long-range dependencies and contextual relationships

- Attention-based pooling:
  - Learns weights over time steps, allowing the model to focus on the most emotionally informative parts of the speech signal

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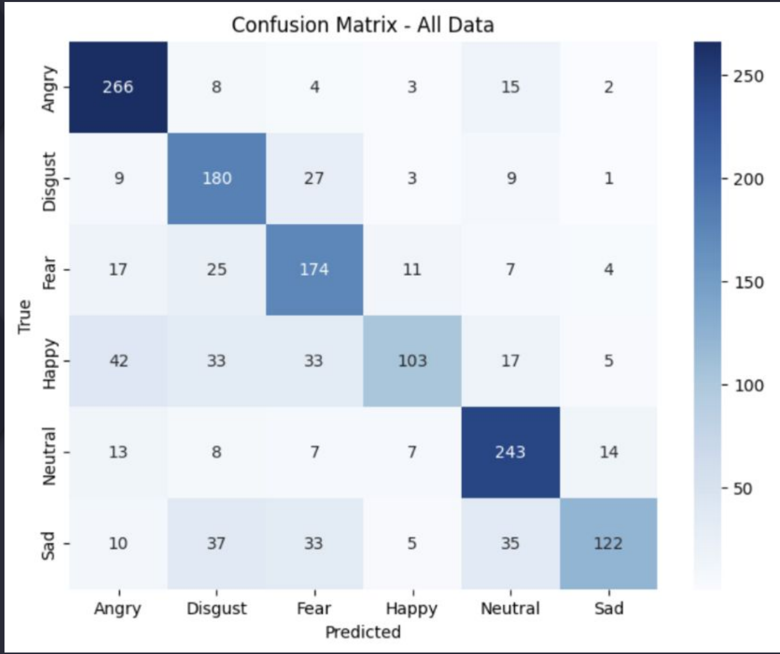
Result

Performance Summary

- Test Accuracy: **71%**

Emotion	Precision	Recall	F1-score	Support
Angry	0.75	0.89	0.81	298
Disgust	0.62	0.79	0.69	229
Fear	0.63	0.73	0.67	238
Happy	0.78	0.44	0.56	233
Neutral	0.75	0.83	0.79	292
Sad	0.82	0.50	0.63	242

Emotion-Level Results



- Strongest performance on angry and neutral
- Disgust and fear share overlapping acoustic features, which can cause confusion
- Lower performance is observed for sad and particularly happy

Model struggles with subtle or low-intensity emotions, where acoustic differences are less pronounced

Dataset-Level Results

- The model performs very well on clean, structured datasets like TESS and RAVDESS Song. But performance drops significantly on real-world datasets like Friends (33.3%)
- EMOVO (50.0%) presented moderate difficulty, which could be caused by differences in language (Italian) and less consistent intensity

★ The model performs best on clean and well-structured datasets. In contrast, performance degrades on datasets with greater variability, naturalistic speech, and less distinct emotional boundaries

Overall

- Emotion recognition performance is not only emotion-dependent but also highly dataset-dependent

- Emotions with clear and strong acoustic signatures are more reliably classified, while those with subtle or overlapping characteristics remain challenging—especially in datasets with greater linguistic diversity or naturalistic recording conditions

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Conclusion

In Conclusion

- 👉 It is possible to extract useful emotional representations from heterogeneous audio data
- 👉 Emotions with clear acoustic patterns, such as anger and neutral, are more reliably recognized, while more subtle emotions like happiness and sadness remain difficult due to overlapping features
- 👉 Dataset characteristics—such as recording quality, speaker diversity, and expression style—do have a significant impact on performance

Limitations + Future Work

- Limitations:
  - The datasets vary in quality and labeling standards, which makes learning harder
  - The model requires substantial training time on a standard laptop GPU, making extensive hyperparameter tuning difficult
- Future:
  - Improving data quality and consistency
  - Integrating additional modalities, such as text or facial expressions

References

- Liu, Zhuang, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. "A convnet for the 2020s." In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 11976-11986. 2022.

Thank You!